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Editor-in-Chief
SoftwareX

Dear Editor,

Please find enclosed our manuscript, “**weatherisk: A Python Package for Spatial Clustering of Climate Extremes via Max-Stable Processes**”, which we are submitting for consideration in *SoftwareX*.

The work grew out of a practical need we kept running into during our own research: applying spatial-extremes methods to gridded climate output requires chaining together several technically demanding steps—trend removal, block-maxima extraction, GEV fitting, Fréchet transformation, local max-stable parameter estimation, and finally clustering—and there was no single Python tool that handled all of them. The R implementation of Contzen, Dickhaus, and Lohmann (“Regionalization approaches for the spatial analysis of extremal dependence”, *Extremes*, 2025, <https://doi.org/10.1007/s10687-025-00519-2>) exists and is well-validated, but porting results back and forth between R and Python for a large HPC run is cumbersome. We wrote **weatherisk** to close that gap.

The package (MIT licence, v0.1.1; archived at <https://doi.org/10.5281/zenodo.20108563>) exposes the full pipeline through both a command-line interface and a Python API. Two clustering approaches are supported—Local Estimates Clustering (LEC) and Extremal Dependence Clustering (EDC)—and every numerical step is cross-checked against the original R code to make sure results are consistent across implementations. As a concrete demonstration, we revisit a published precipitation clustering study for the AWI-ESM-1-1-LR model (192×96 grid, 1850–2005), running the full pipeline in Python on a high-performance cluster and obtaining a spatially consistent cluster structure that is broadly comparable to the published result.

The manuscript follows the SoftwareX author guidelines. It contains the mandatory metadata table, descriptions of the three implemented pipeline flows with worked CLI examples, two result figures, and a code appendix with self-contained snippets for each flow. The source code is publicly available on GitHub and permanently archived on Zenodo.

W. Kchikach built the software, ran all experiments, and wrote the paper. T. Dickhaus and G. Lohmann guided the scientific direction and reviewed the manuscript. All authors have read the final version and agree to this submission. We have no competing interests to declare.

We hope **weatherisk** will be useful to researchers working at the intersection of spatial statistics and climate science, and we look forward to the reviewers’ comments.

Yours sincerely,

Widad Kchikach
(on behalf of all co-authors)